Adaptive Force Field Evaluation

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Outline

1. Summary of Previous Discussions
2. Description of System's Moving Parts
3. Experiments

Diffusion and Flow Models

This discussion was mainly aimed at pinning down the math behind diffusion and flow models. During the whiteboard discussion, the following points were discussed at length

- Ordinary Differential Equations- Existence and Uniqueness theorems
- Random Processes with focus on Brownian Motion
- Idea of generation as sampling from a distribution
- Conditional and Marginal probabillity paths

Problem Definition for Coding Interview

- **Given:** A set of n charges and their locations in $\mathbb{R}^3$ and a dataset of m points and corresponding force fields due to the given charges.
- **Goal:** Circumvent the O(n) computations required for force field evaluation at a new point using the given dataset.
- **Proposed Solution:**
  1. Train a **neural network** using the dataset to output force fields.
  2. Evaluate **confidence** on neural network prediction using **Conformal Prediction**
  3. If Confidence $\leq \alpha$ (a user defined threshold), fall back to brute force computation.
  4. Otherwise output neural network prediction.

Force Field Predictor NN

- Composed of **four ResNet blocks**
- Each ResNet block is composed of **two linear layers with swish activation function**
- Trained using Huber loss.

Conformal Prediction - Theory

- Framework for quantification of uncertainty on model's output
- Gives statistical guarantees without any assumptions on underlying data distribution.
- Only requires the data points to be **IID**

Conformal Prediction - Theory

The following workflow needs to be implemented for input x and output y

1. Identify heuristc uncertainty notion from pre trained model
2. Define score function $s(x, y) \in \mathbb{R}$ (Larger the score, worse the prediction)
3. $\hat{q} = \frac{\lceil (n+1)(1-\alpha) \rceil}{n}$ quantile of calibration scores $s_1 = s(x_1, y_1), \cdots s_n = s(x_n, y_n)$
4. Prediction sets for test point: $C(x_{test}) = \{y : s(x_{test}, y) \leq \hat{q}\}$

Conformal Prediction - Theory

Theorem(Conformal Coverage Guarantee): Suppose $(X_i, Y_i)_{i=1}^n$ and (X_{test}, Y_{test}) are iid and define $\hat{q}$ and $C(X_{test})$ as in previous slide then the following holds

$$P(Y_{test} \in C(X_{test})) \geq 1 - \alpha$$

Conformal Prediction - Implementation

1. Denote by $\hat{f}(x)$, our force field predictor NN
2. Train a NN $\sigma(x)$ to predict $|y - \hat{f}(x)|$ for a location x and corresponding force field y.
3. Define $s(x, y) = \frac{|y - \hat{f}(x)|}{\sigma(x)}$
4. Implement the workflow with score function $s(x, y)$
5. We will allow the user to define a **maximum error** in addition to a confidence threshold so that we can calculate the confidence for the worst case scenario.

Conformal Prediction Implementation

The error predictor neural network $\sigma(x)$ is a **three layer neural network with Swish activations** and is trained using Huber loss.

List of Experiments

1. **Convergence Analysis:** To monitor the training of neural nets.
2. **Scaling Laws:** How does change in dataset size affect the fall back ratio.
3. **Spatial Confidence Heatmap:** A visualization of model's confidence in space.
4. **Parameter Sensitivity:** How does maximum error and confidence threshold affect fall back ratio.
5. **Adversarial Stress Test:** Attempt to artificially lower model's confidence and trigger a fall back.

For all the experiments we will fix the number of charges to 100.

Training on 30K points- Convergence Analysis

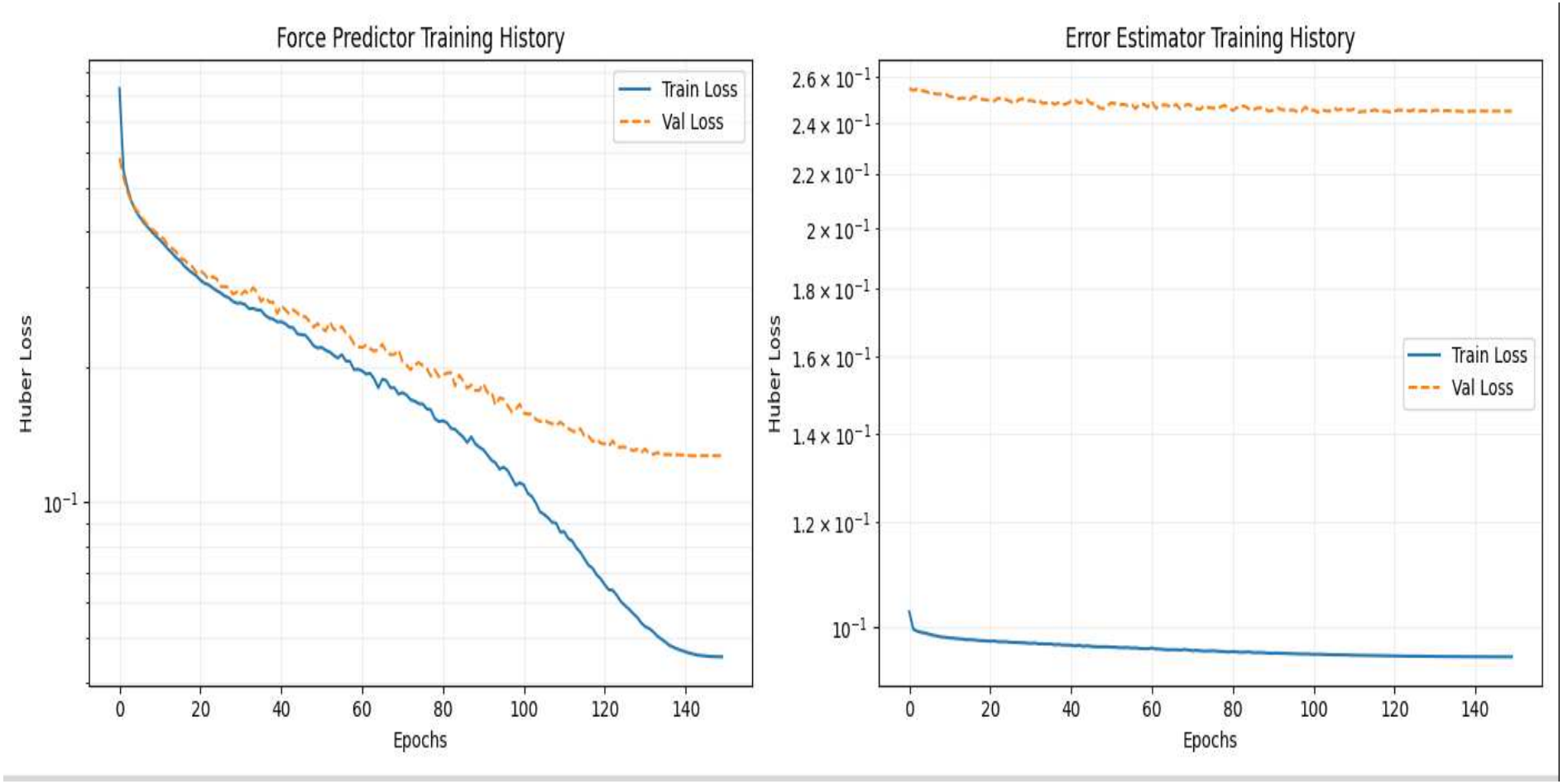


Figure: Charges are configured to be in a spherical shell of 2-11

Training on 30K points- Convergence Analysis

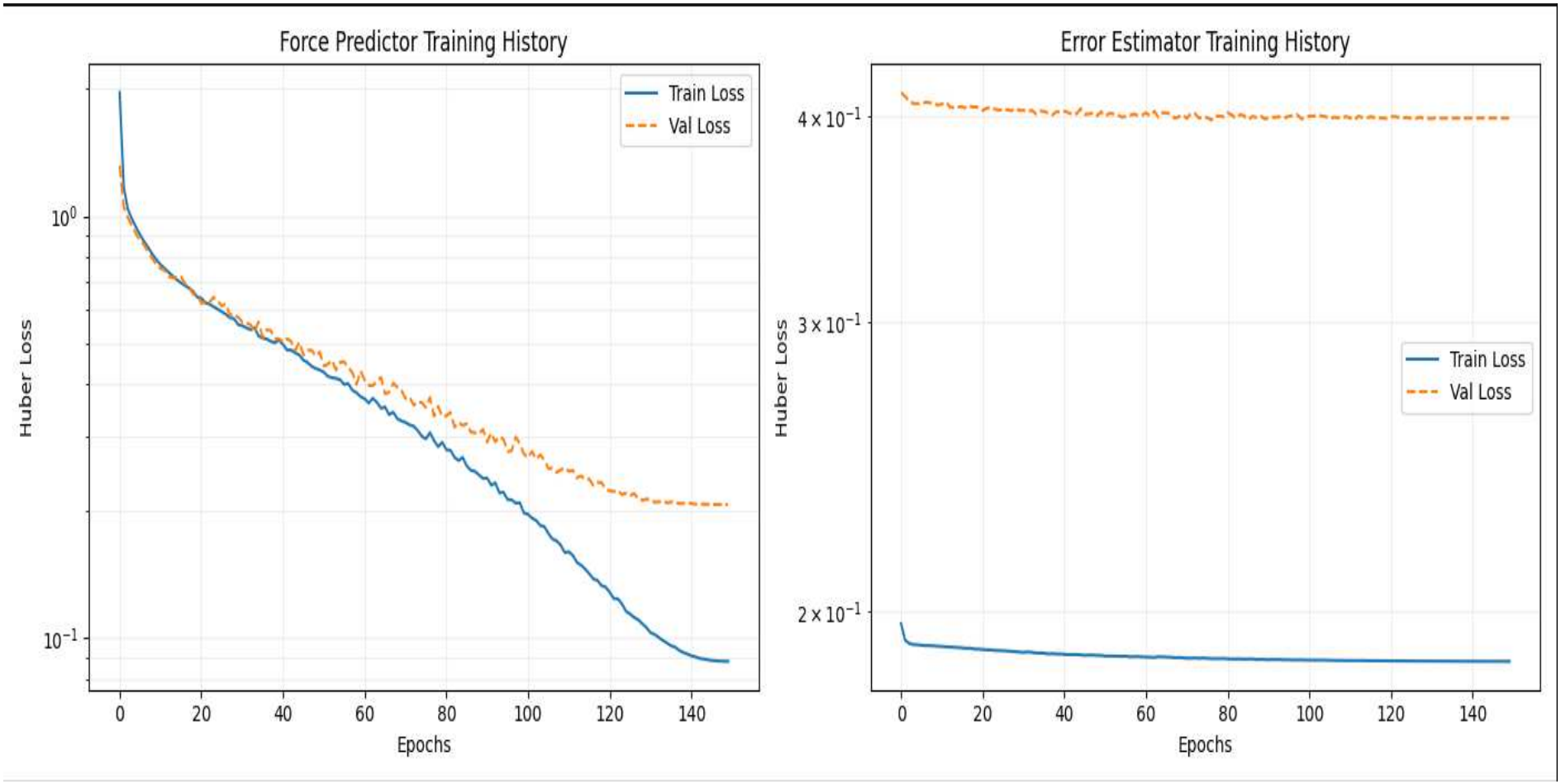


Figure: Charges are configured to be in a spherical shell of 2-3

Training on 30K points- Overall Performance

For charges configured to be in a spherical shell of 2-11:

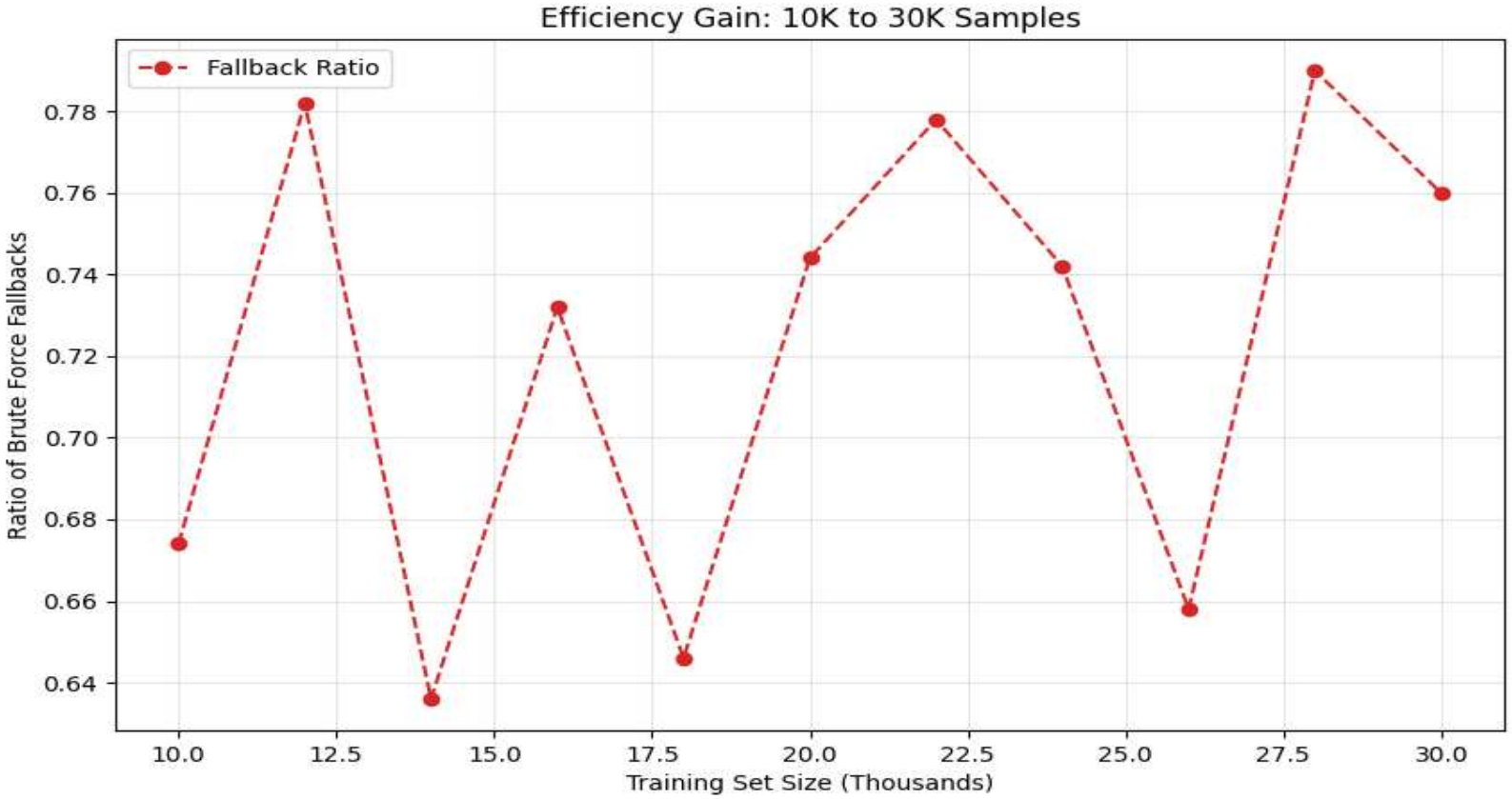
- **Fall Back Percentage:** 71.54%
- **Avg Time Per Query:** 0.0072 ms

Training on 30K points- Overall Performance

For charges configured to be in a spherical shell of 2-3:

- **Fall Back Percentage:** 15.43%
- **Avg Time Per Query:** 0.0036 ms

Training on 30K points - Scaling Laws



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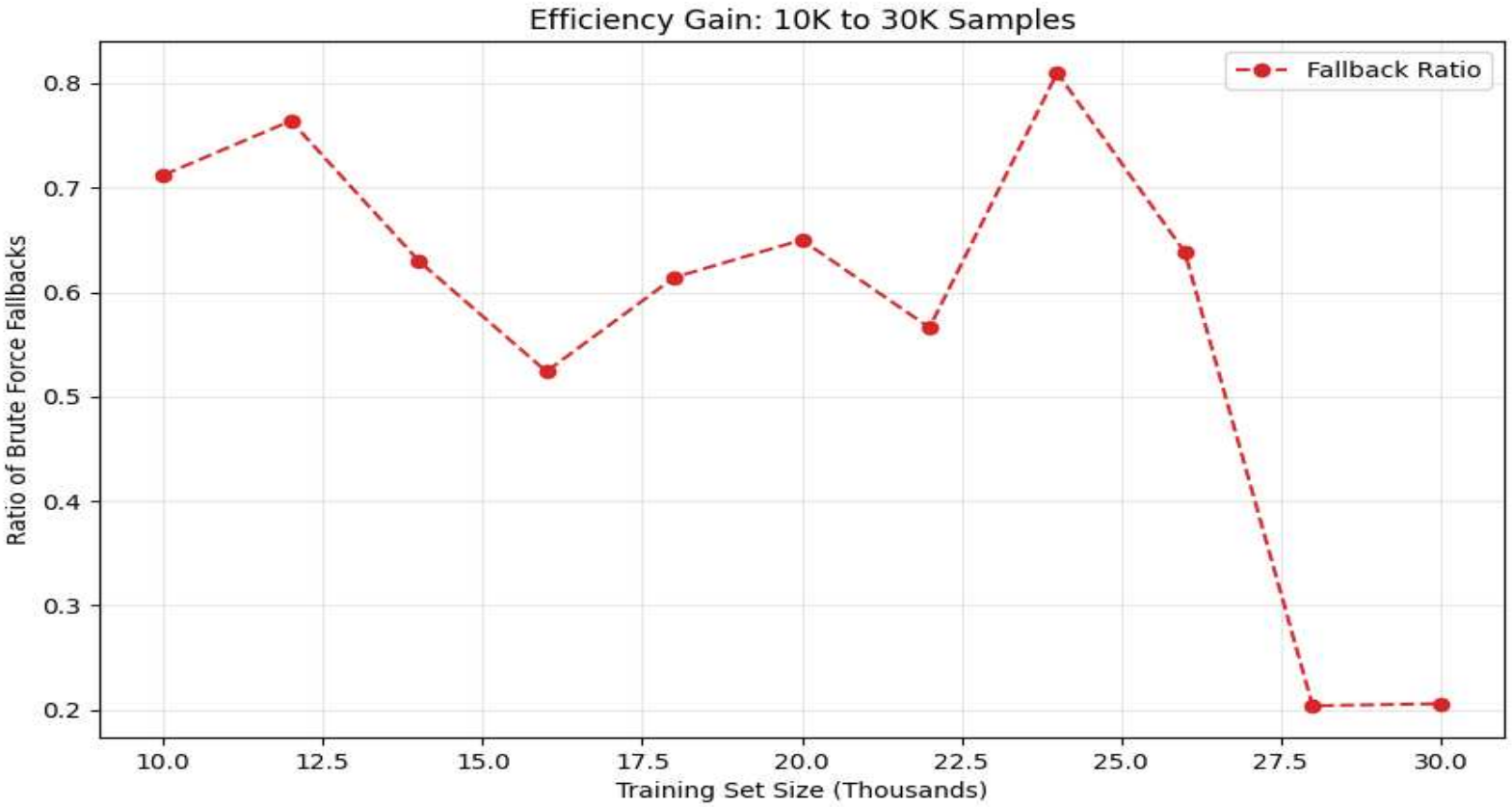

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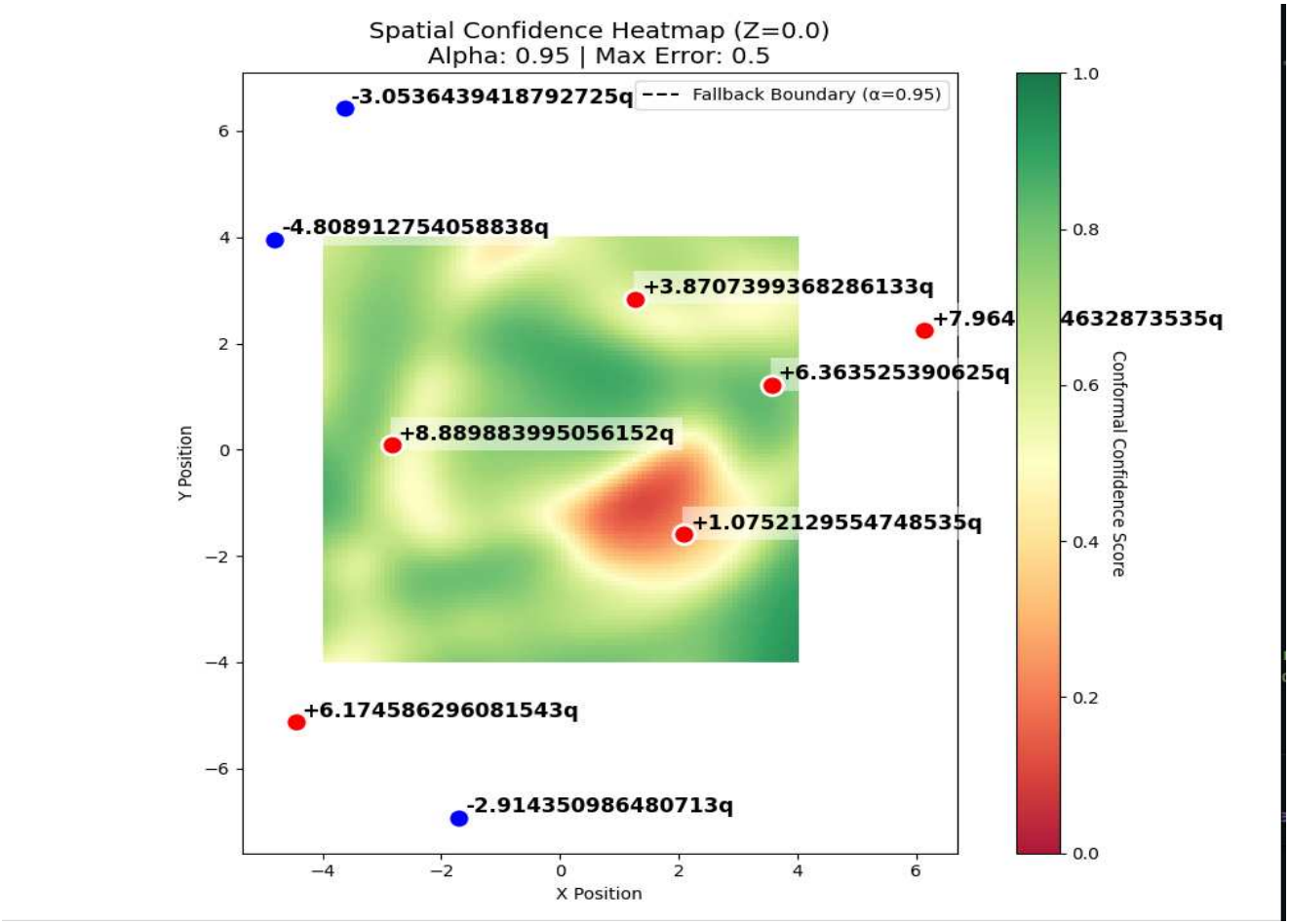
Training on 30K points - Confidence Heatmap



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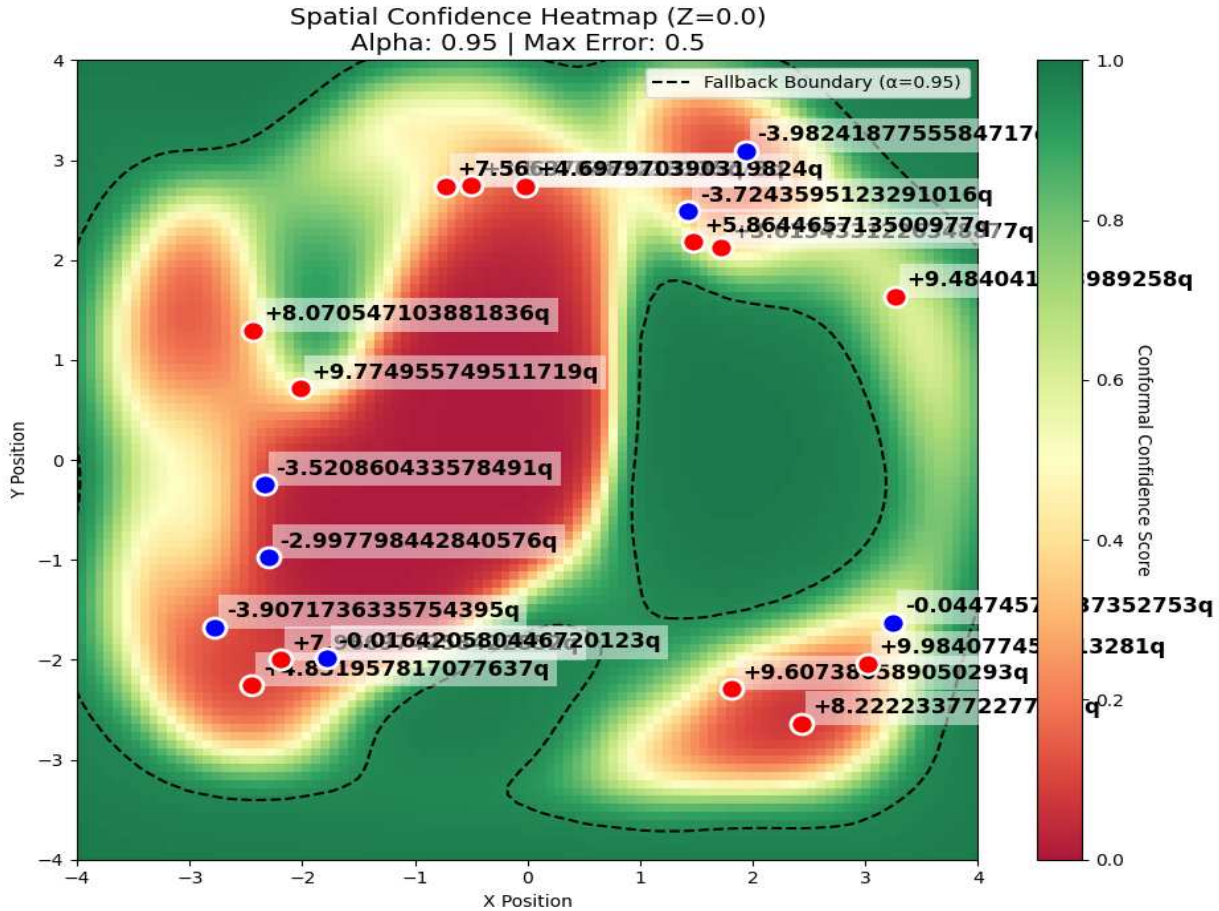

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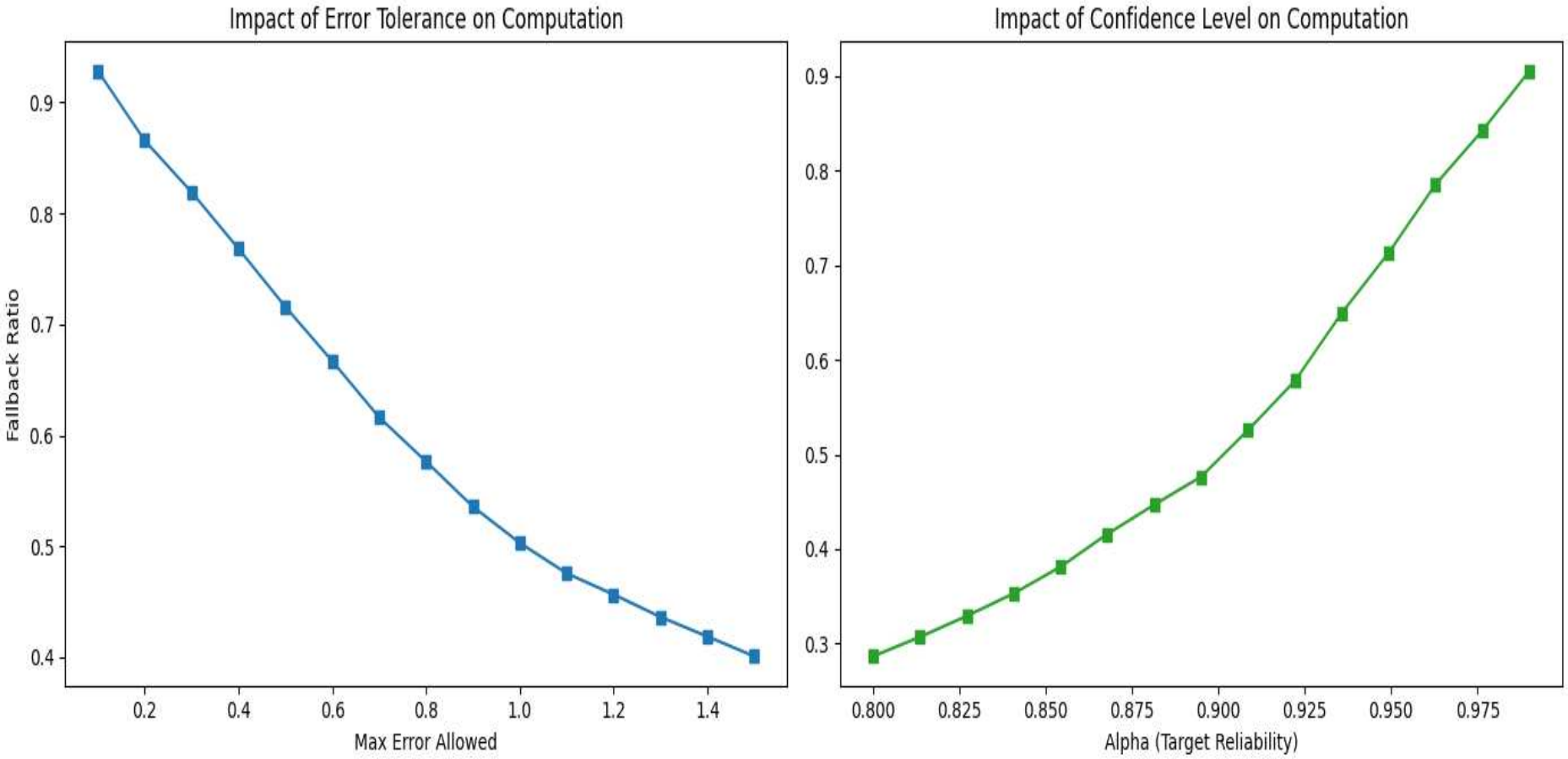
Training on 30K points - Parameter Sensitivity



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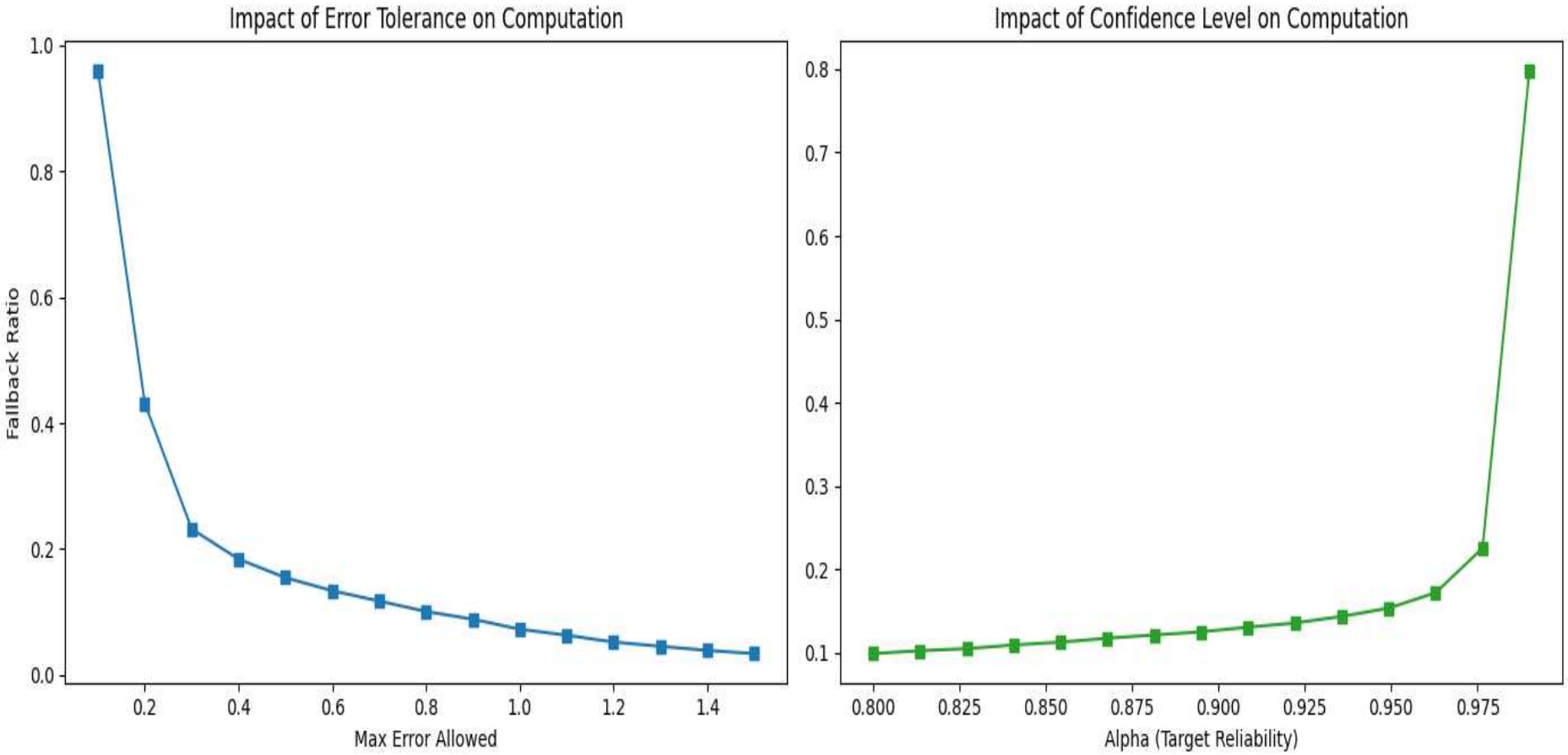

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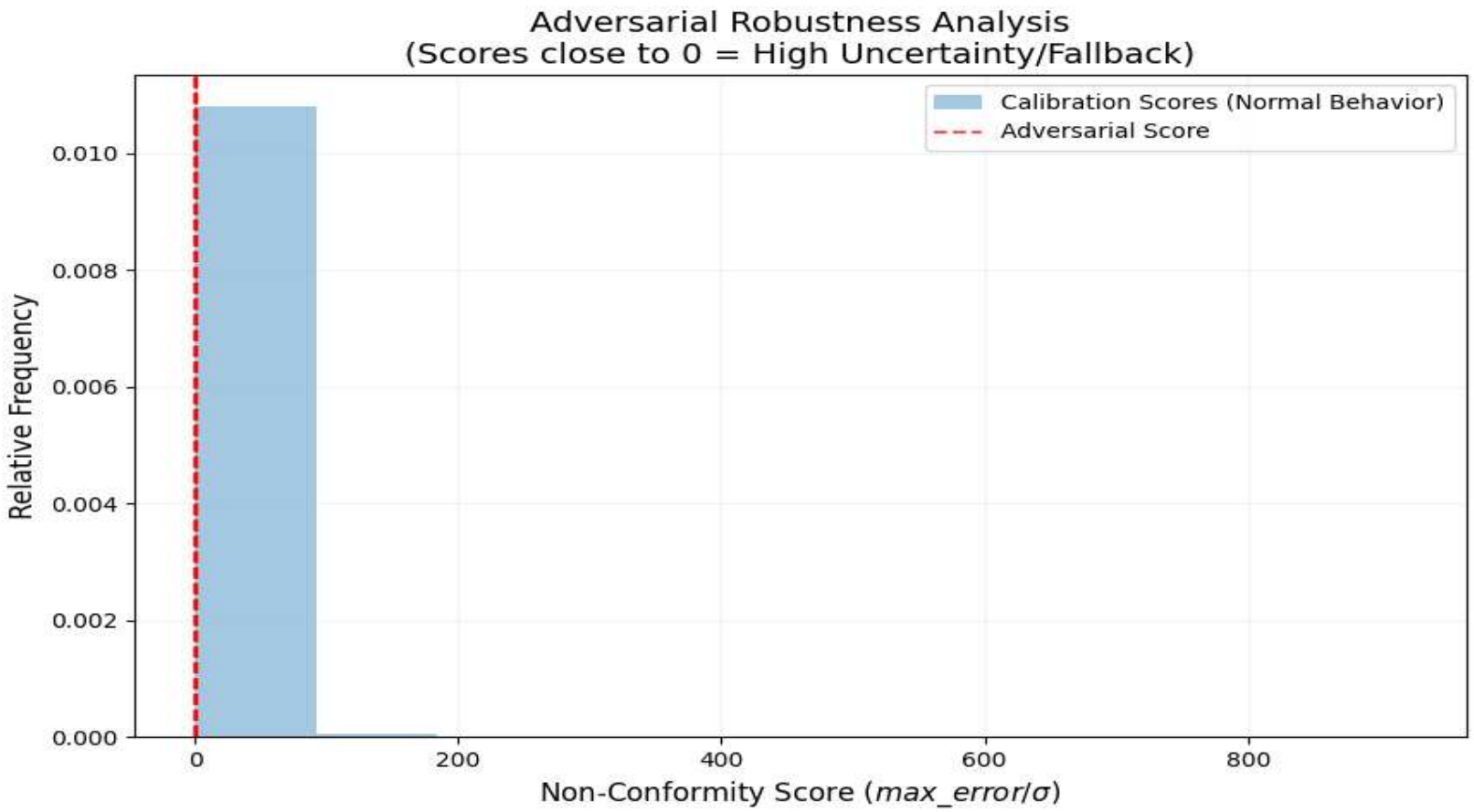
Training on 30K points - Adversarial Stress Test



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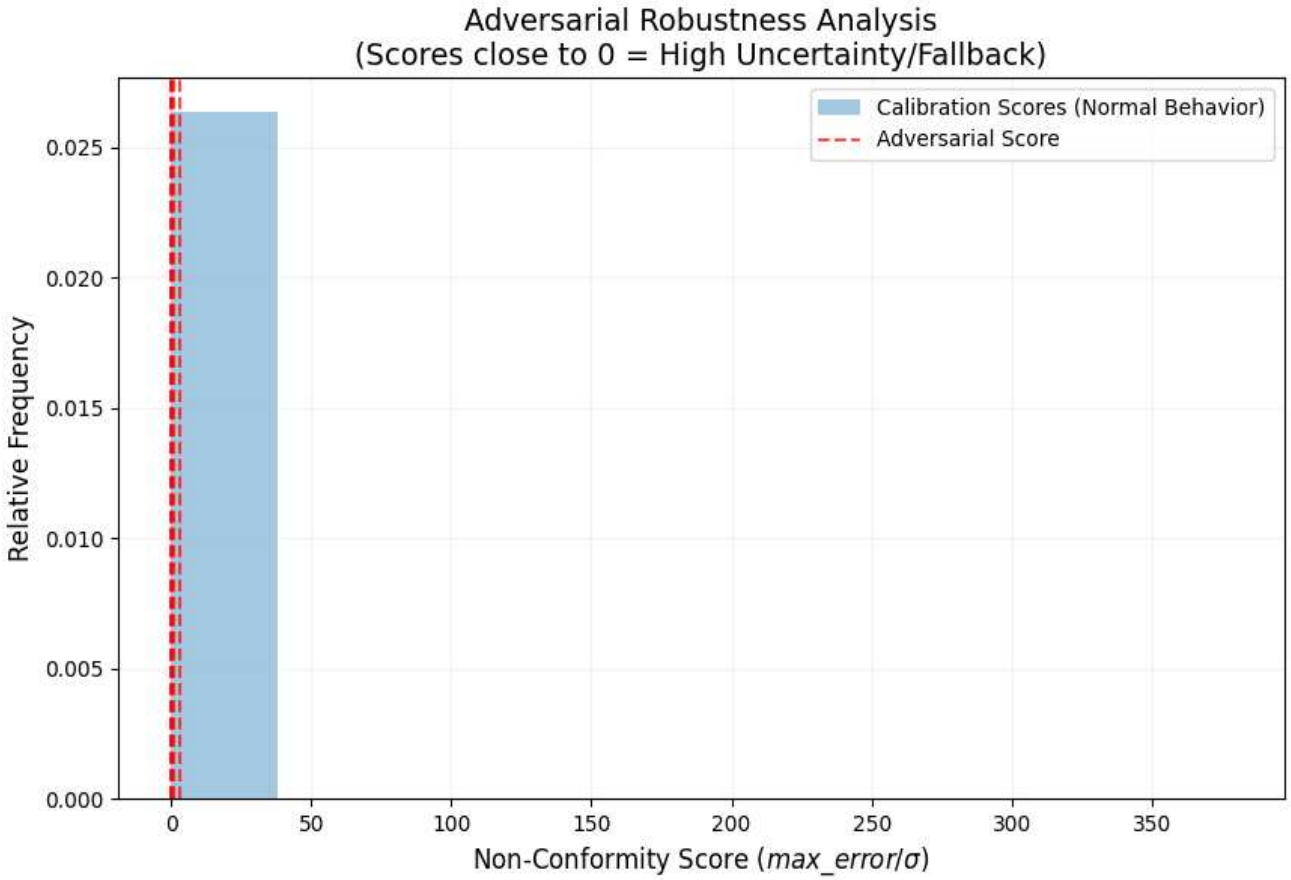

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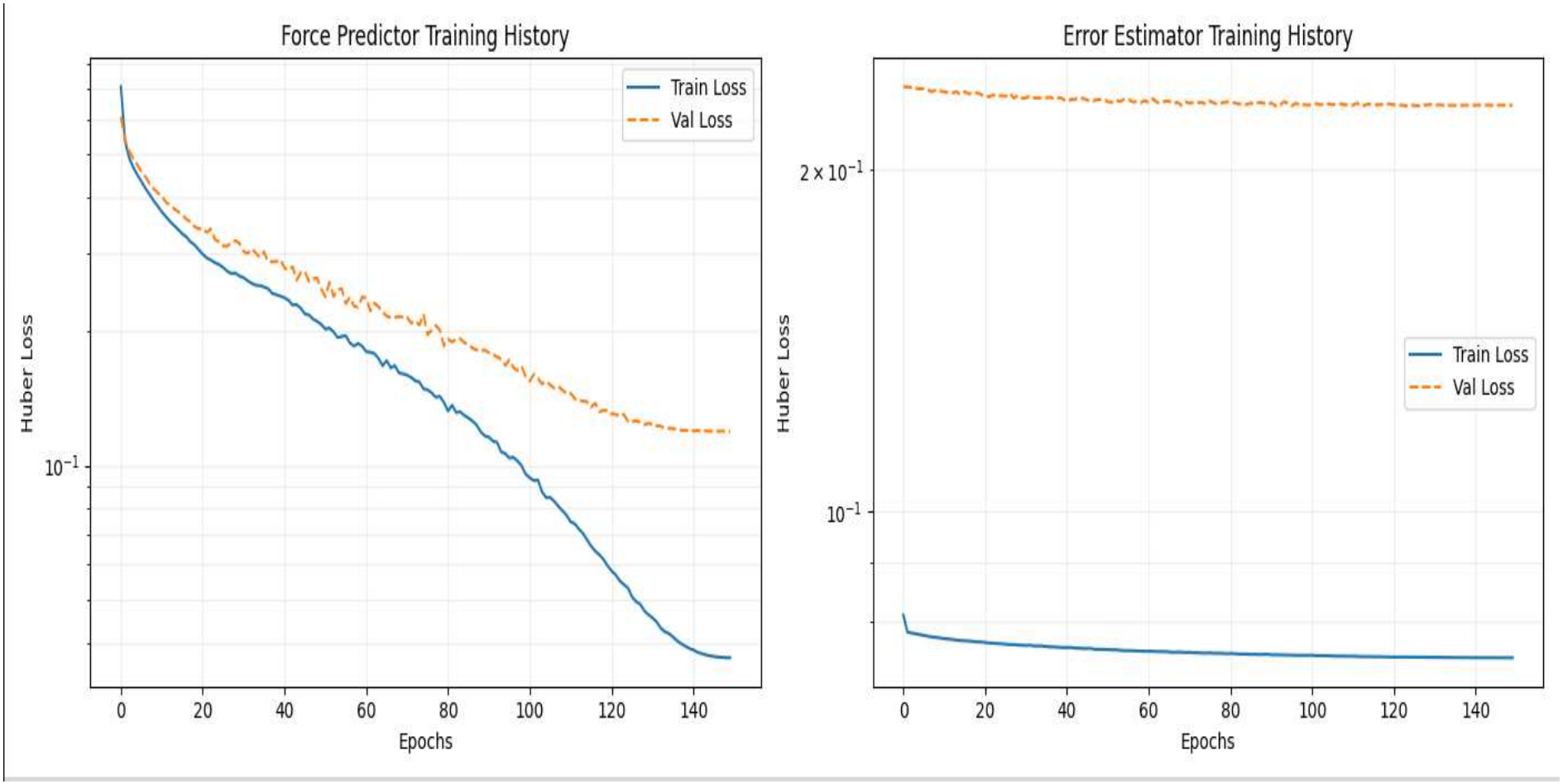
Training on 60K points- Convergence Analysis



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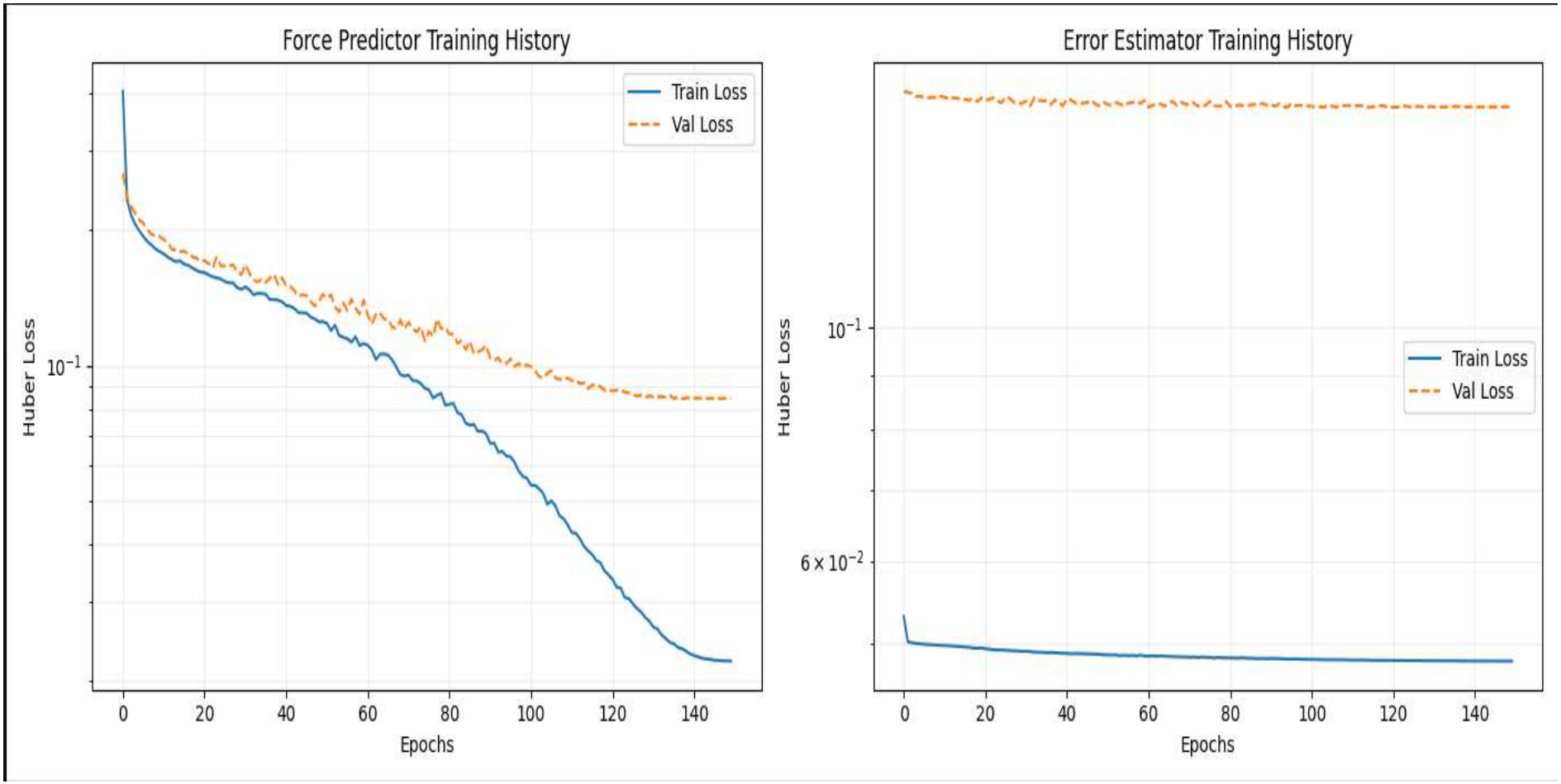

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Training on 60K points- Overall Performance

For charges configured to be in a spherical shell of 2-11:

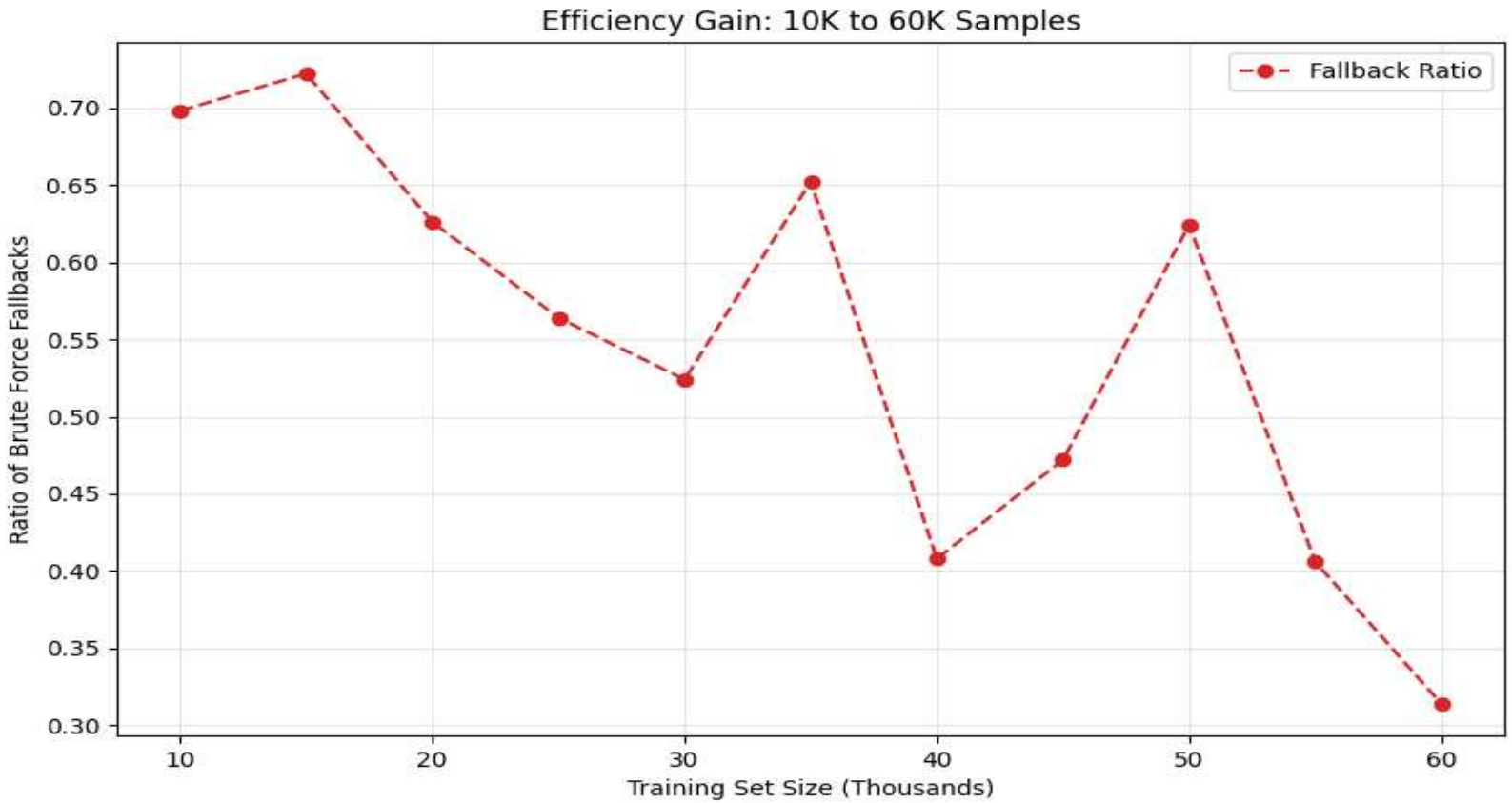
- **Fall Back Percentage:** 30%
- **Avg Time Per Query:** 0.0052 ms

Training on 60K points- Overall Performance

For charges configured to be in a spherical shell of 2-3:

- **Fall Back Percentage:** 5%
- **Avg Time Per Query:** 0.0042 ms

Training on 60K points- Scaling Laws



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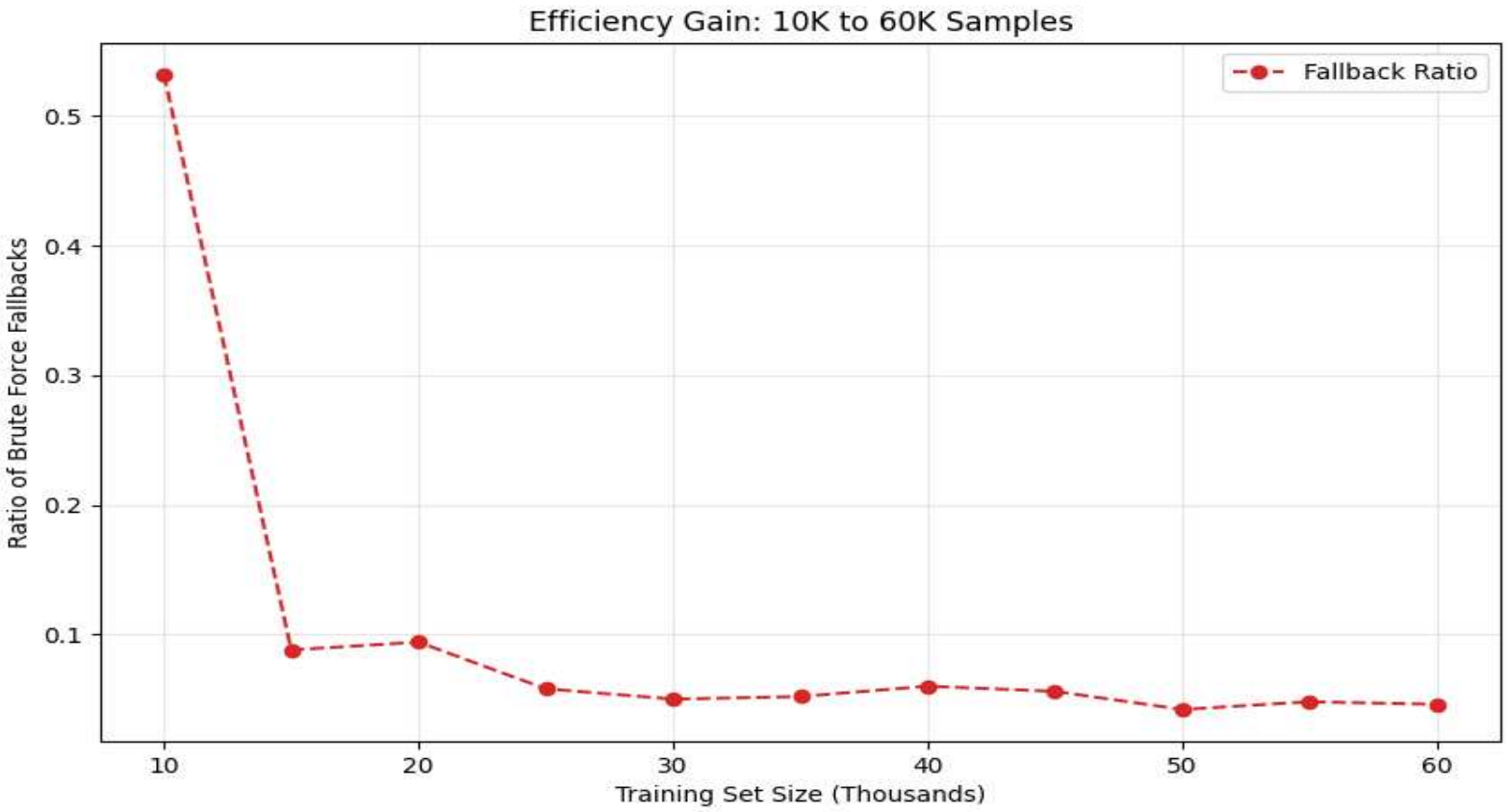

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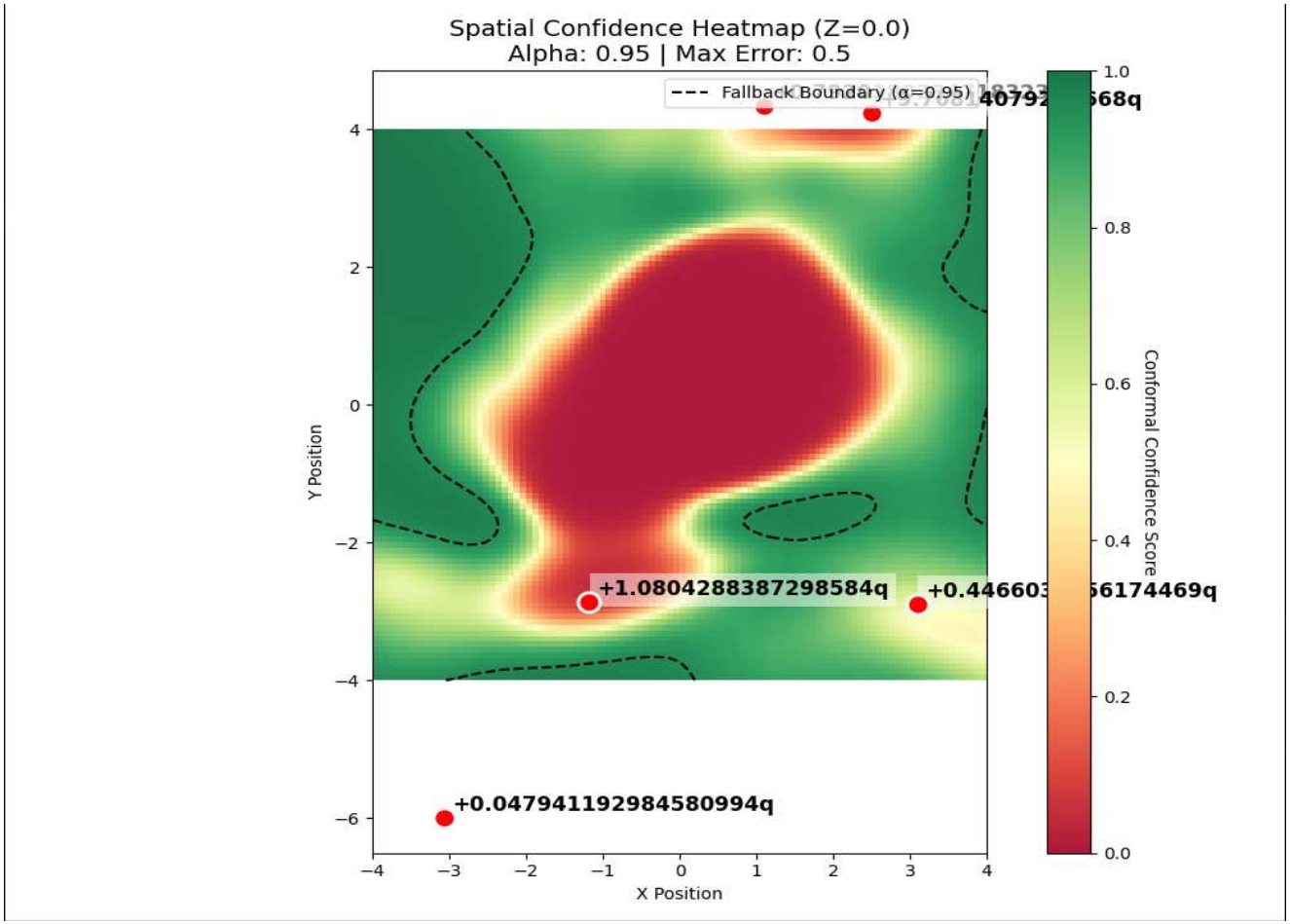

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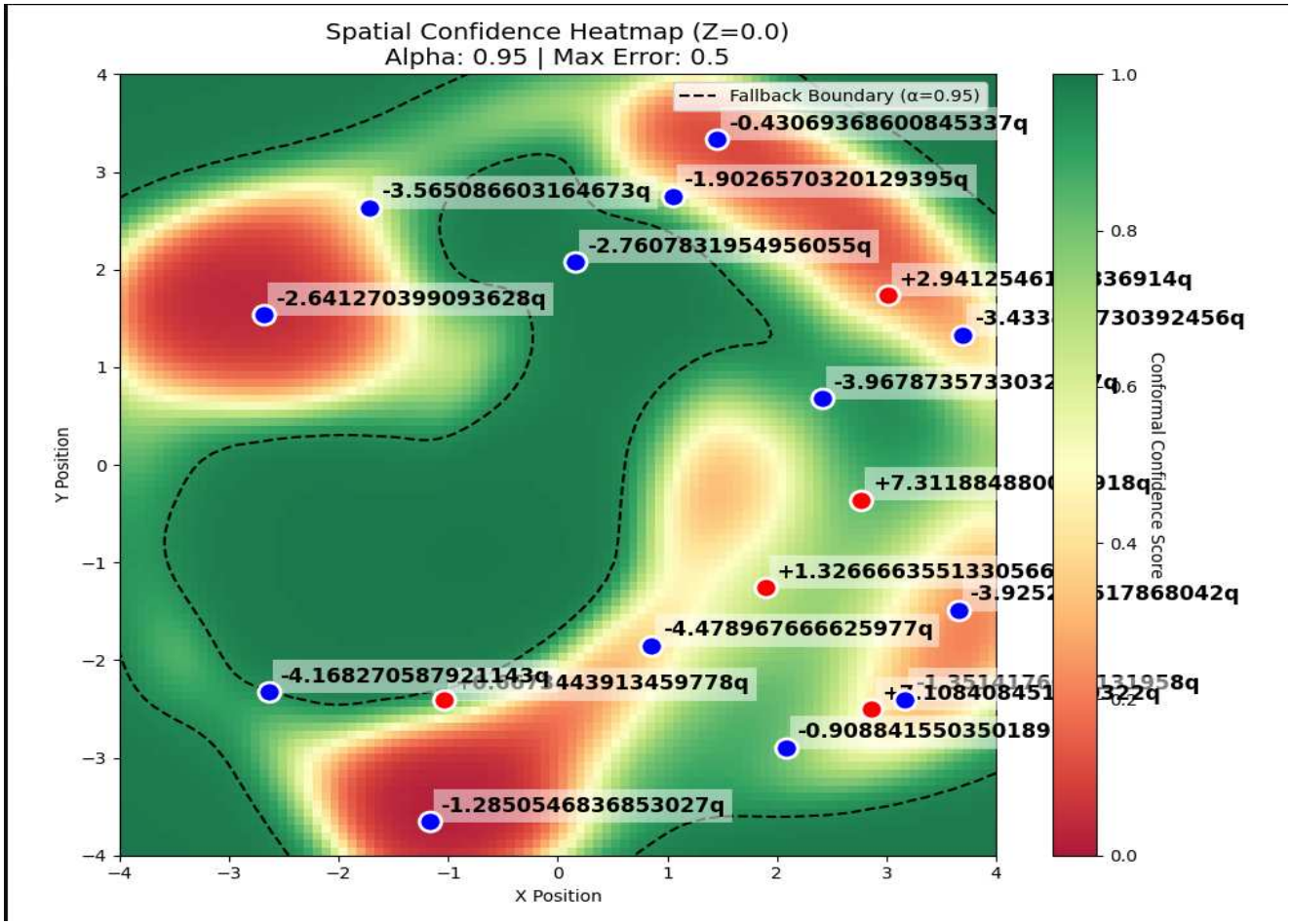

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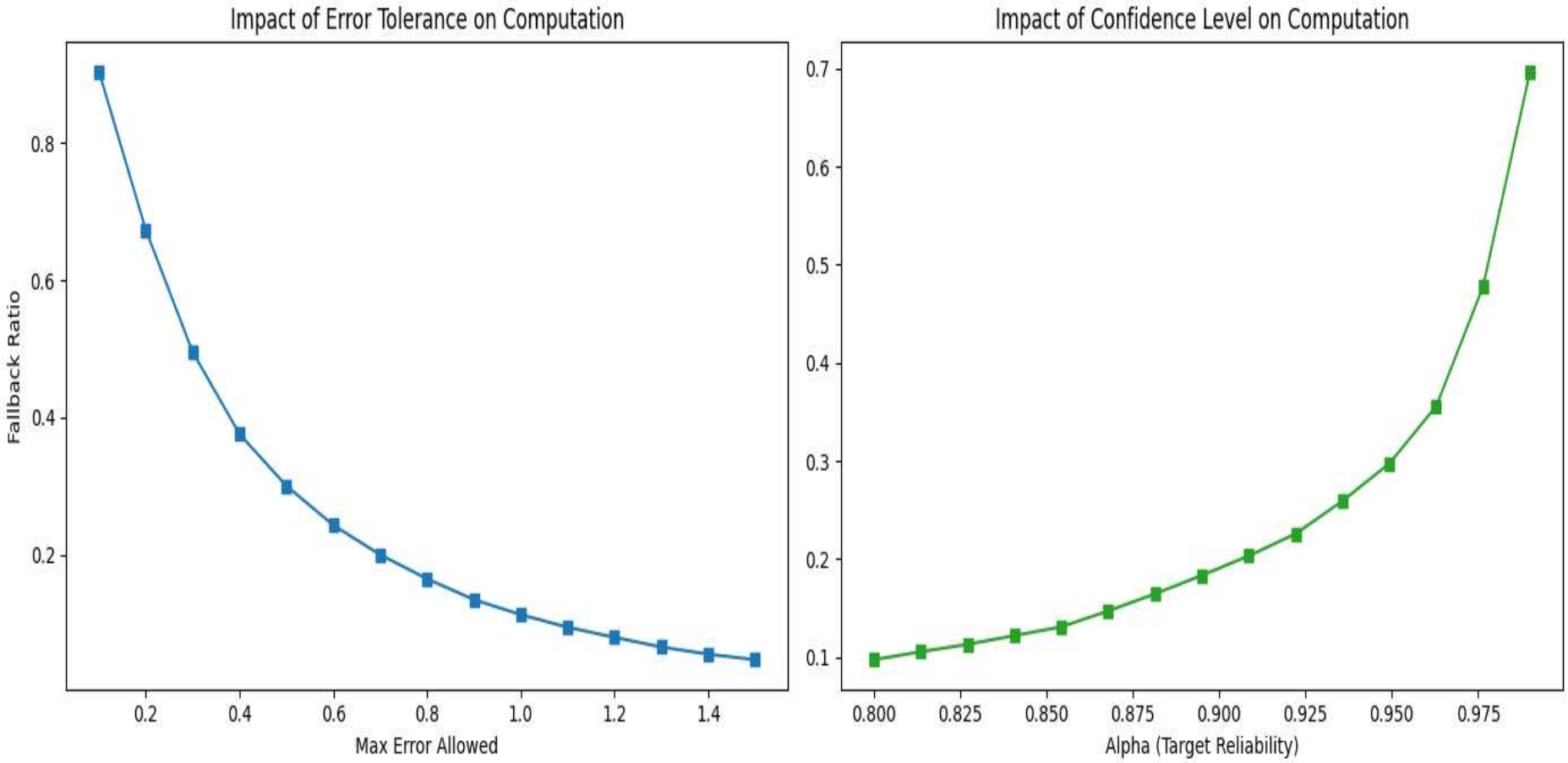
Training on 60K points- Parameter Sensitivity



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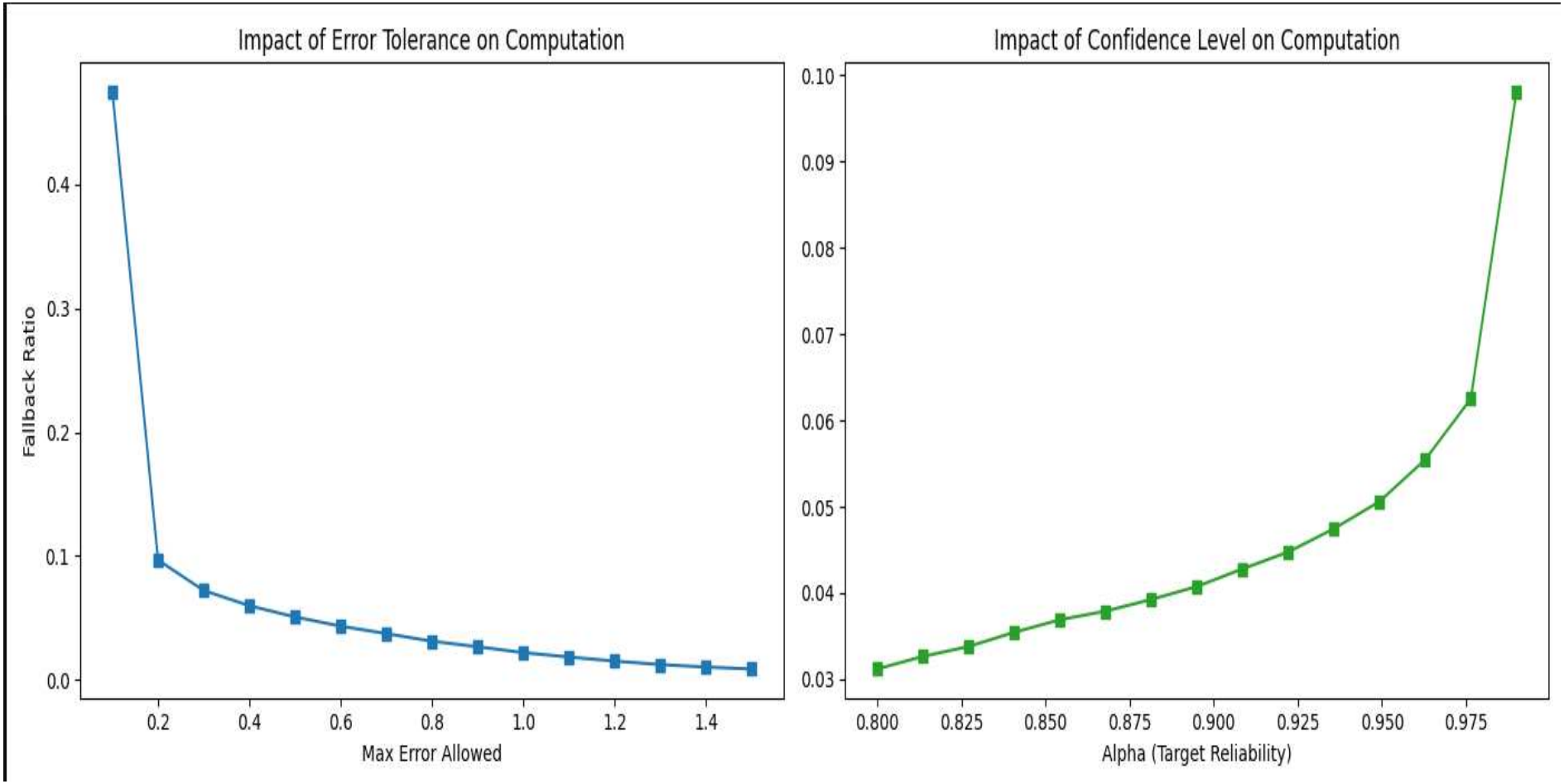

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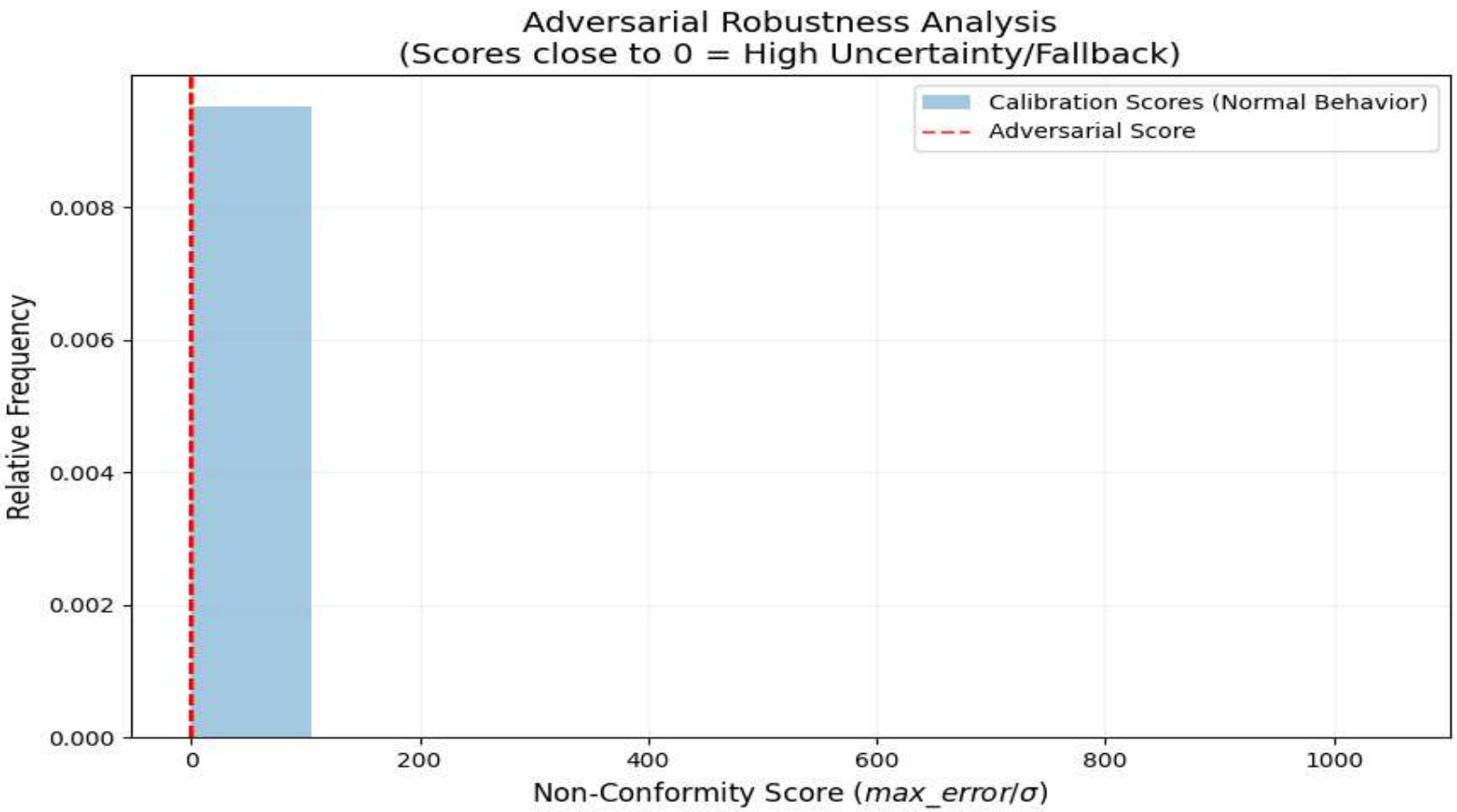

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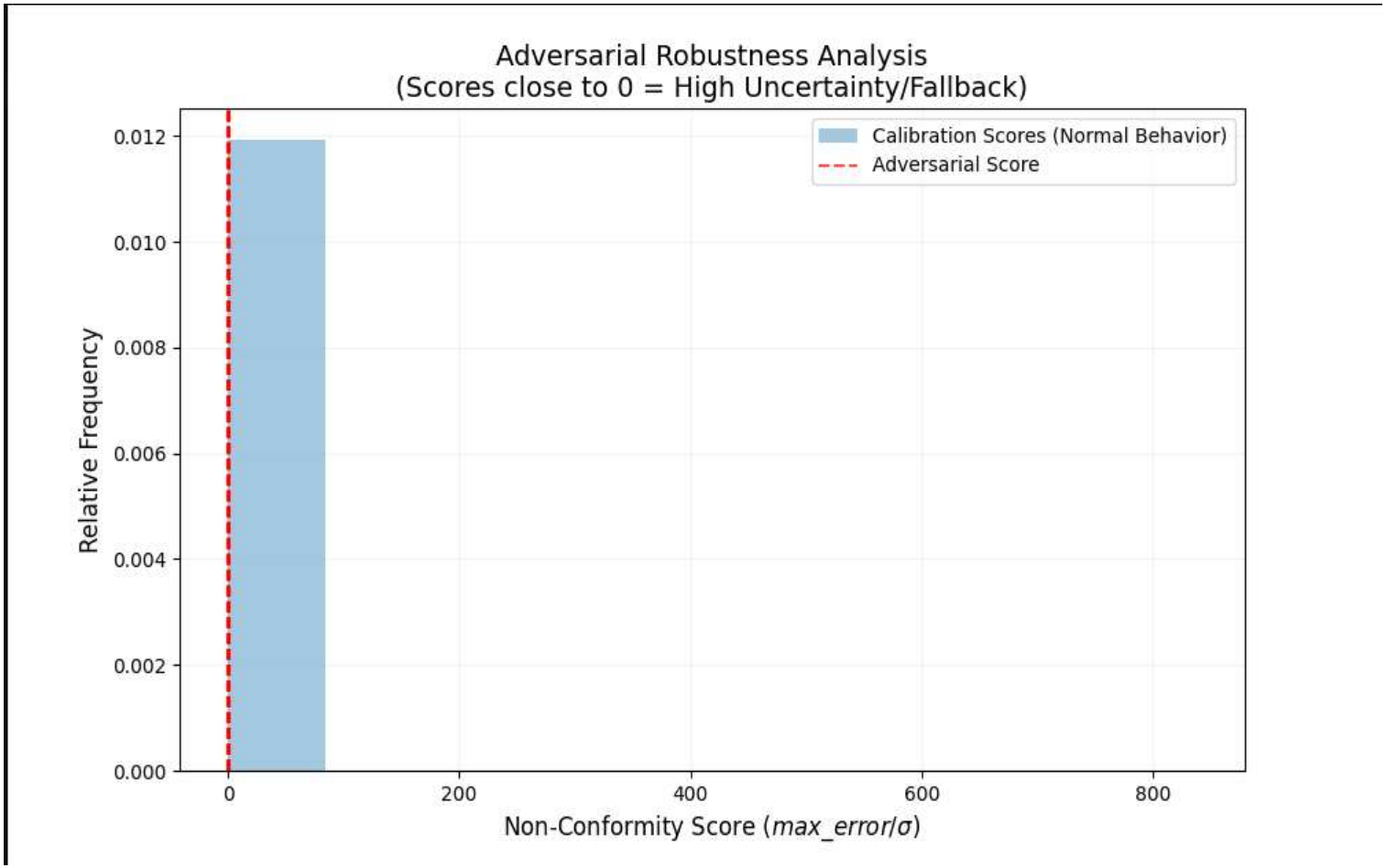

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Conclusions

- Fallback ratio tends to be low when charges are clustered together.
- Confidence thereshold of 90% results in fairly low fall back ratio.
- Neural network is less confident on space near charges.
- Fallback ratio fluctuates highly as a function of dataset size but overall performance is better for bigger dataset.
- System is computationally vulnerable to adversarial attacks but it correctly outputs low confidence due to statistical guarantees on conformal prediction.

Thank You!

Questions or Comments?